

5.2.2 Voronoi Neighborhood Averaging (VNA) for Monitor Data

If you choose the *Voronoi Neighborhood Averaging* option, BenMAP-CE first identifies the set of monitors that "surround" each grid cell's center (these monitors are referred to as the grid cell's neighbors), and then BenMAP-CE calculates an inverse-distance weighted average of these neighboring monitors. In this section, we provide some examples of the different ways that BenMAP-CE calculates the average of the neighboring monitors. See Appendix B on Air Pollution Exposure Estimation Algorithms for an expanded discussion of VNA, including how the VNA algorithm actually chooses the neighbor monitors, as well as the different ways that it may be used.

VNA interpolation has three advanced interpolation options, which can be modified by clicking on the **Advanced** button at the bottom of the window:

- **Maximum Neighbor Distance (in km)** specifies the maximum distance that a monitor may be from the center of a grid cell, and still be included in the set of neighbor monitors used to calculate air pollution exposure at a particular grid cell. The default setting is infinite (i.e., no limit to the monitor's distance to the center of the grid cell).
- **Maximum Relative Neighbor Distance** specifies the maximum ratio for the distance of each included monitor to the distance of the closest monitor. The default setting is infinite.
- **Weighting Approach** specifies whether BenMAP-CE should use inverse-distance weighting for the monitors, or inverse-distance-squared weighting of the monitors. The default setting is inverse-distance weighting.

The following examples illustrate how varying these options affects the final average concentration estimate.

Example 1: Monitor Data VNA method

Default options

Consider the following example at a hypothetical rural grid cell, where there are relatively few monitors, and where the distance from a monitor to the center of a grid cell can be fairly large. With *VNA*, BenMAP-CE first identifies the set of "neighbor" monitors for each grid cell. The number of neighbors is usually in the range of about three to eight. In this case, assume that there are five monitors at distances of 25, 50, 100, 200, and 400 km from the grid cell, with annual PM_{2.5} levels of 8, 13, 12, 18, and 15 µg/m³, respectively. BenMAP-CE would calculate an inverse-distance weighted average of the monitor values as follows:

$$PM_{2.5} \text{ average} = \frac{\frac{1}{25} \bullet 8 + \frac{1}{50} \bullet 13 + \frac{1}{100} \bullet 12 + \frac{1}{200} \bullet 18 + \frac{1}{400} \bullet 15}{\frac{1}{25} + \frac{1}{50} + \frac{1}{100} + \frac{1}{200} + \frac{1}{400}} = 10.68$$

Example 2: Monitor Data VNA method

Maximum Neighbor Distance = 75

Using the same example that we used above, let us say you have specified a **Maximum Neighbor Distance** of 75 km, and left unchanged the default options (infinite value) for **Maximum Relative Neighbor Distance**. BenMAP-CE would only consider the first two monitors, and would calculate an inverse-distance weighted average of the monitor values as follows:

$$PM_{2.5} \text{ average} = \frac{\frac{1}{25} \bullet 8 + \frac{1}{50} \bullet 13}{\frac{1}{25} + \frac{1}{50}} = 9.67$$

Example 3: Monitor Data VNA method

Maximum Relative Neighbor Distance = 10

Alternatively, if you left the **Maximum Neighbor Distance** at the default (infinite), but have set the **Maximum Relative Neighbor Distance** to 10, then BenMAP-CE would calculate the ratio of the distance for each monitor to distance of the closest monitor. In this case, the ratios would be 1 (=25/25), 2 (=50/25), 4 (=100/25), 8 (=200/25), and 16 (=400/25), and BenMAP-CE would drop the monitor with a ratio of 16. BenMAP-CE would then calculate an inverse-distance weighted average of the monitor values as follows:

$$PM_{2.5} \text{ average} = \frac{\frac{1}{25} \bullet 8 + \frac{1}{50} \bullet 13 + \frac{1}{100} \bullet 12 + \frac{1}{200} \bullet 18}{\frac{1}{25} + \frac{1}{50} + \frac{1}{100} + \frac{1}{200}} = 10.53$$

Example 4: Monitor Data VNA method

Inverse-distance squared neighbor weighting

In addition, you can specify an inverse-distance-squared weighting of the monitors. Let us say that you have left unchanged the defaults (infinite values) for **Maximum Neighbor Distance** and **Maximum Relative Neighbor Distance**, and specified that the

Weighting Approach is *Inverse Distance Squared*. BenMAP-CE would then calculate an inverse-distance-squared weighted average of the monitor values as follows:

$$PM_{2.5} \text{ average} = \frac{\frac{1}{625} \cdot 8 + \frac{1}{2,500} \cdot 13 + \frac{1}{10,000} \cdot 12 + \frac{1}{40,000} \cdot 18 + \frac{1}{160,000} \cdot 15}{\frac{1}{625} + \frac{1}{2,500} + \frac{1}{10,000} + \frac{1}{40,000} + \frac{1}{160,000}} = 9.26$$

Example 5: Monitor Data VNA method

Maximum Neighbor Distance = 75

Maximum Relative Neighbor Distance = 10

Inverse-distance squared Weighting Approach

Finally, you could specify changes to all three options: a **Maximum Neighbor Distance** of 75 km, a **Maximum Relative Neighbor Distance** of 10, and a **Weighting Approach** of *Inverse Distance Squared* weighting. BenMAP-CE would then calculate the following average:

$$PM_{2.5} \text{ average} = \frac{\frac{1}{625} \cdot 8 + \frac{1}{2,500} \cdot 13}{\frac{1}{625} + \frac{1}{2,500}} = 9.00$$

5.2.3 Fixed Radius for Monitor Data

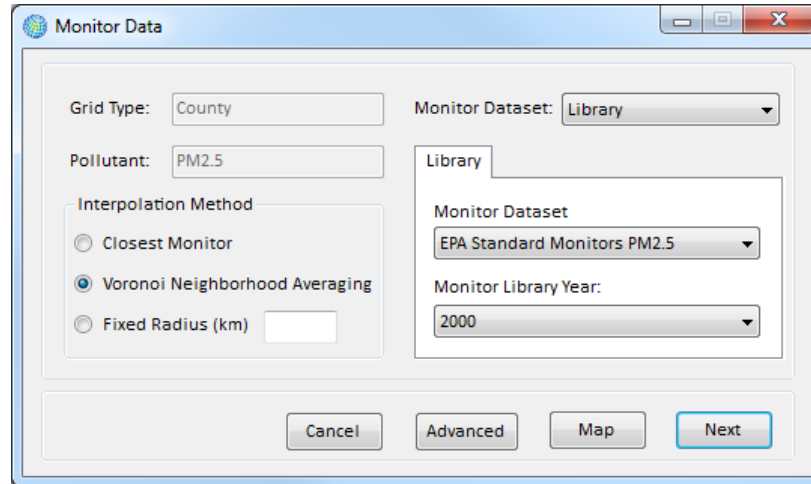
If you choose the *Fixed Radius (km)* option, BenMAP-CE averages all of the monitor values within a fixed radius (measured in kilometers) that you specify. The way that the monitor values are averaged depends on the **Weighting Approach** that you choose after clicking the **Advanced** button. You can choose either *Inverse Distance* or *Inverse Distance Squared* weighting.

Note that the default option with the *Fixed Radius* approach is that BenMAP-CE will not calculate an average for a grid cell if there are no monitors within the fixed radius (distance) that you specify. In the **Advanced Options** window, if you select *Get Closest if None within Radius*, then for those grid cells without any monitors within the fixed radius, BenMAP-CE will choose the nearest monitor (regardless of distance) and apply that value as the “average”.

5.2.4 Custom Monitor Filtering for EPA Standard Monitors

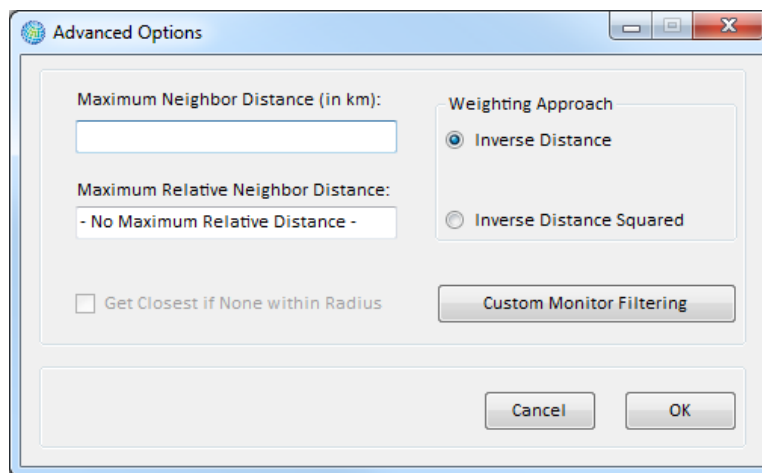
Custom Monitoring Filtering options apply only to the EPA Standard Monitors library in the United States setup; these are the only monitoring values that include all of the variables that BenMAP-CE needs in order to filter the data properly. This tool allows you to filter, map, and export your monitor data. You can reach the **Custom Monitor**

Filtering tool by first choosing your pollutant, data source (e.g., monitor library) and year on the **Monitor Data** window.



The **Monitor Data** dialog box is shown. It has a title bar with a globe icon and standard window controls. The main area contains several fields and a list. On the left, there are three sections: 'Grid Type' with a dropdown set to 'County', 'Pollutant' with a dropdown set to 'PM2.5', and 'Interpolation Method' with three radio buttons: 'Closest Monitor', 'Voronoi Neighborhood Averaging' (which is selected), and 'Fixed Radius (km)' with an empty text box. On the right, there is a 'Monitor Dataset' dropdown set to 'Library'. Below this is a list box titled 'Library' containing 'EPA Standard Monitors PM2.5'. Below the list box is a 'Monitor Library Year' dropdown set to '2000'. At the bottom, there are four buttons: 'Cancel', 'Advanced', 'Map', and 'Next'.

Click the **Advanced** button. This will take you to the **Advanced Options** window.



The **Advanced Options** dialog box is shown. It has a title bar with a globe icon and standard window controls. The main area contains several fields and a list. On the left, there are three sections: 'Maximum Neighbor Distance (in km):' with an empty text box, 'Maximum Relative Neighbor Distance:' with a dropdown set to '- No Maximum Relative Distance -', and a checkbox labeled 'Get Closest if None within Radius' which is unchecked. On the right, there is a 'Weighting Approach' section with two radio buttons: 'Inverse Distance' (which is selected) and 'Inverse Distance Squared'. Below this is a button labeled 'Custom Monitor Filtering'. At the bottom, there are two buttons: 'Cancel' and 'OK'.

Click the **Custom Monitor Filtering** button. This will take you to the **Filter Monitors** window.